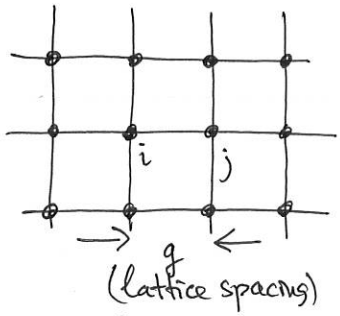
Where field theory is coming from?

Study <u>Ising model</u>: defined on the lattice

classical variables $\sigma_r = \pm 1$ (spins)



$$H = \sum_{ij} J_{ij}\, \sigma_i \sigma_j$$

let's choose $J_{ij} = \begin{cases} -J_0 & \text{for } \langle i,j \rangle \text{ nearest neighbors} \\ 0 & \text{otherwise} \end{cases}$

Partition function $Z = \sum_{\{\sigma_n = \pm 1\}} e^{-\beta H} = \sum_{\{\sigma_n = \pm 1\}} e^{\sum_{\langle ij \rangle} \beta J_0 \sigma_i \sigma_j}$ $\left(\beta = \frac{1}{T}\right)$

sum over all possible spin configurations

Z depends on dimensionality and lattice geometry.

Exact solutions: (1) for $d=1$ chain (easy)

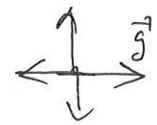
(2) for $d=2$ square lattice (very hard. done by Lars Onsager in 1944)

(3) $d=3$: no solution yet.

We use this popular model to derive Landau-Ginzburg action and motivate approximate treatment which can later be extended to other, more complicated, problems.

We do this with the help of Hubbard-Stratonovich transformation.

$$\mathcal{H} = \sum_{i,\vec{g}} \left(-\frac{1}{2} J_0\right) \sigma_i \, \sigma_{i+\vec{g}}$$



$$= \sum_{i,\vec{g}} \left(-\frac{1}{2} J_0\right) \frac{1}{N} \sum_{k_1 k_2} e^{i\vec{k}_1 \vec{r}_i + i\vec{k}_2(\vec{r}_i + \vec{g})} \sigma_{k_1} \sigma_{k_2}$$

$$= -\frac{1}{2} J_0 \sum_{\vec{k}\vec{g}} e^{-i\vec{k}\vec{g}} \sigma_k \sigma_{-k} = \sum_{\vec{k}} -J(\vec{k}) \sigma_k \sigma_{-k}$$

where $J(\vec{k}) = \frac{1}{2} \sum_{\vec{g}} J_0 \, e^{-i\vec{k}\cdot\vec{g}}$

$$Z = \sum_{(\sigma)} \exp \sum_{\vec{k}} \beta J(k) \sigma_k \sigma_{-k}$$

<u>Hubbard-Stratonovich:</u>

Consider $I = \int d\varphi_k \; e^{\left(-\frac{a(k)}{2} \varphi_k \varphi_{-k} + \varphi_k y_{-k} + \varphi_{-k} y_k\right)}$

$$-\frac{a(k)}{2}\left(\varphi_k \varphi_{-k} - \frac{2}{a(k)} \varphi_k y_{-k} - \frac{2}{a(k)} \varphi_{-k} y_k\right)$$

$$= -\frac{a(k)}{2}\left\{\left(\varphi_k - \frac{2}{a(k)} y_k\right)\left(\varphi_{-k} - \frac{2}{a(+k)} y_{-k}\right) - \frac{4}{a_k^2} y_k y_{-k}\right\}$$

$$a(k) = a(-k).$$

<u>So y_k is σ_k, and $\frac{2}{a_k} = \beta J(k)$:</u>

$$I = \int d\varphi_k \; \exp -\frac{a(k)}{2}\left(\varphi_k - \frac{2 y_k}{a_k}\right)\left(\varphi_{-k} - \frac{2 y_{-k}}{a_k}\right) + \frac{2}{a_k} y_k y_{-k}$$

$$= (\text{cust}) \; e^{\frac{2}{a_k} y_k y_{-k}} \Rightarrow \frac{2}{a_k} = \beta J(k)$$

$$Z = \sum_{\sigma} \int d\varphi_k \; e^{\sum_k \left(-\frac{1}{\beta J(k)} \varphi_k \varphi_{-k} + \varphi_k \sigma_{-k} + \varphi_{-k} \sigma_k\right)}$$

$$\sum_k \varphi_k \sigma_{-k} = \sum_{\vec{k}} \frac{1}{N} \sum_{\vec{r}_1 \vec{r}_2} e^{-i\vec{k}\cdot\vec{r}_1} \varphi(\vec{r}_1) \, e^{i\vec{k}\cdot\vec{r}_2} \sigma(\vec{r}_2)$$

$$= \sum_r \varphi_r \sigma_r$$

Re-scale $\varphi_k \to \sqrt{\beta J(k)}\,\varphi_k$

$$Z = \int d\varphi_k \; e^{\sum_k (-\varphi_k \varphi_{-k})} \; \underbrace{e^{\sum_r 2\tilde{\varphi}_r \sigma_r}}$$

where $\tilde{\varphi}_r = \frac{1}{\sqrt{N}} \sum \qquad \exp \ln ch(2\tilde{\varphi}_r)$

$$\sum_k \tilde{\varphi}_k \sigma_{-k} = \sum_k \tilde{\varphi}_{-k} \sigma_k = \sum_k \tilde{\varphi}_{-k} \frac{1}{\sqrt{N}} \sum_r e^{i\vec{k}\cdot\vec{r}} \sigma_r$$

$$= \sum_r \sigma_r \left(\frac{1}{\sqrt{N}} \sum_k \varphi_{-k} \sqrt{\beta J(k)}\, e^{i\vec{k}\cdot\vec{r}} \right) = \tilde{\varphi}_r$$

$\Rightarrow$ we'll get $\quad +\frac{1}{2}(2\tilde{\varphi}_r)^2 - \frac{1}{12}(2\tilde{\varphi}_r)^4$

$$\Rightarrow \sum_r 2(\tilde{\varphi}_r)^2 = 2\sum_r \frac{1}{N} \sum_{k_1 k_2} \varphi_{-k_1} \varphi_{-k_2} \sqrt{\beta^2 J(k_1) J(k_2)}\, e^{i(\vec{k}_1+\vec{k}_2)\vec{r}}$$

$$= 2\sum_k \varphi_k \varphi_{-k} \beta J(k)$$

$$Z = \int d\varphi_k \; \exp \sum_k (2\beta J(k) - 1)\, \varphi_k \varphi_{-k} - \frac{4}{12} \sum_r (\tilde{\varphi}_r)^4$$

$$= \int d\varphi_k \; \exp - \beta \left[\sum_k \left(-2J(k) + \frac{1}{\beta}\right) \varphi_k \varphi_{-k} + \frac{4}{3\beta} \sum_r (\tilde{\varphi}_r)^4 \right]$$

$$J(k) = J_0 \left(\cos(k_x g) + \cos(k_y g) + \cos(k_z g) \right)$$

$$\approx J_0 \left(3 - \frac{1}{2} (\vec{k}\cdot\vec{g})^2 \right) = 3J_0 \left(1 - \frac{g^2}{6}(k_x^2 + k_y^2 + k_z^2) \right)$$

$$= \int d\varphi_k \; \exp - \beta \sum_k \left(-6J_0 + J_0 g^2 \vec{k}^2 + \frac{1}{\beta} \right) \varphi_k \varphi_{-k} + \frac{4}{3\beta} \sum_r (\tilde{\varphi}_r)^4$$

Now transform to real space

$$\sum_{\vec{k}} \varphi_{-k}\, \vec{k}^2\, \varphi_k = \frac{1}{N} \sum_{\vec{k}} \sum_{r_1 r_2} \varphi(r_2)\, e^{-i\vec{k}\cdot\vec{r}_2}\, \underset{\substack{\| \\ k_x^2+k_y^2+k_z^2}}{\vec{k}^2}\, e^{i\vec{k}\cdot\vec{r}_1}\, \varphi(r_1) =$$

$$k_x^2\, e^{i\vec{k}\cdot\vec{r}_1} = k_x^2\, e^{i(k_x x_1 + k_y y_1 + k_z z_1)} =$$

$$= -\frac{\partial^2}{\partial x_1^2}\, e^{i(\vec{k}-\vec{r}_1)}$$

$$= \sum_r \varphi(\vec{r})\left(-\vec{\nabla}_r^2\right)\varphi(\vec{r})$$

$$Z = \int D\varphi(\vec{r})\, \exp - \sum_{\vec{r}} \varphi(r)\left[1 - 6J_0\beta - \beta J_0 g^2\, \vec{\nabla}_r^2\right]\varphi(r) + \frac{4}{3}(\beta J_0)^2\, \varphi^4(r)$$

$$\sum_r g^3 \{\ldots\} \to \int d\vec{r}\, \{\quad\}$$

volume of unit cell

Rescale field $\varphi_r \to \frac{g^{3/2}}{\sqrt{2}}\, \varphi(\vec{r})$

$$Z = \int D\varphi(\vec{r})\, \exp - \int d\vec{r} \left\{ \frac{1}{2}\rho\left(\vec{\nabla}\varphi(r)\right)^2 + \frac{1}{2} m^2 \varphi^2(r) + \frac{1}{4} u\, \varphi^4(r) \right\}$$

where $\rho = \beta J_0$ (dimensionless)

$$m^2 = 1 - 6J_0\beta = \frac{T - 6J_0}{T} \cong \frac{T - T_c}{T_c}, \quad T_c = 6J_0$$

* Obtained partition function as path integral over configurations of classical real field $\varphi(\vec{r})$ = <u>order parameter</u>

$$Z = \int D\varphi(\vec{r})\, e^{-S_{LG}} = \int D\varphi\, e^{-\int d^d\vec{r}\, \mathcal{L}(\varphi)}$$

$$S_{LG} = \int d\vec{r} \left(\frac{1}{2}\rho\left(\partial_r \varphi(r)\right)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u\, \varphi^4 + \ldots \right)$$

Note: it is <u>not</u> of the form $e^{-\beta H_{eff}} \Longrightarrow T = \frac{1}{\beta}$ comes via m^2!